

QUICK REFERENCE: Relations & Functions (1-Page Cheat Sheet)

CARTESIAN PRODUCT ($A \times B$)

Definition: All ordered pairs (a,b) where $a \in A$, $b \in B$

Key Properties:

- Non-commutative: $A \times B \neq B \times A$
- Cardinality: $|A \times B| = |A| \times |B|$
- Empty set: If $A = \emptyset$ or $B = \emptyset$, then $A \times B = \emptyset$
- Equality: $(a,b) = (c,d) \iff a=c$ AND $b=d$

Distributive Laws:

- $A \times (B \cup C) = (A \times B) \cup (A \times C)$
- $A \times (B \cap C) = (A \times B) \cap (A \times C)$
- $A \times (B - C) = (A \times B) - (A \times C)$

RELATIONS ($R \subseteq A \times B$)

Definition: Subset of Cartesian product connecting elements

Domain: All first elements = $\{a \mid (a,b) \in R\}$

Range: All second elements = $\{b \mid (a,b) \in R\}$

Codomain: Target set B

Inverse: $R^{-1} =$

RELATION TYPES (On Set A)

Type	Definition	Test
Reflexive	$\forall a \in A: (a,a) \in R$	Every element paired with itself
Symmetric	$(a,b) \in R \implies (b,a) \in R$	Bidirectional pairs
Transitive	$(a,b), (b,c) \in R \implies (a,c) \in R$	Chain property
Antisymmetric	$(a,b), (b,a) \in R \implies a=b$	No opposite pairs unless same
Equivalence	Reflexive + Symmetric + Transitive	Partitions the set

RELATION REPRESENTATIONS

Matrix ($m \times n$): Entry = 1 if $a_i R b_j$, else 0

DiGraph: Vertices connected by directed edges; self-loops for reflexive pairs

FUNCTIONS ($f: A \rightarrow B$)

Definition: Special relation where each $a \in A$ maps to exactly one $b \in B$

Verification: Relation is function if no input has multiple outputs

```
domain = [x for (x,y) in relation]
is_function = len(domain) == len(set(domain))
```

Terminology:

- **Domain**: Input set A
- **Codomain**: Target set B

- **Range:** Actual outputs $\{f(a) \mid a \in A\} \subseteq B$
- **$f(a)$:** Image of a
- **$f^{-1}(b)$:** Preimage of b

FUNCTION TYPES

Type	Definition	Test
Injective (1-1)	Different inputs $\rightarrow$ different outputs	$f(a_1)=f(a_2) \Rightarrow a_1=a_2$
Surjective (Onto)	Every element in B is mapped to	Range = Codomain
Bijective	Both injective AND surjective	Perfect pairing; f^{-1} exists

Inverse Function:

- f^{-1} exists $\iff f$ is bijective
- If $y = f(x)$, then $x = f^{-1}(y)$
- $f^{-1}(f(x)) = x$; $f(f^{-1}(y)) = y$

COMPOSITION

Definition: $(g \circ f)(x) = g(f(x))$ [apply f first, then g]

Properties:

- NOT commutative: $g \circ f \neq f \circ g$
- Associative: $(h \circ g) \circ f = h \circ (g \circ f)$
- Injectivity preserved: f, g injective $\Rightarrow g \circ f$ injective
- Surjectivity preserved: f, g surjective $\Rightarrow g \circ f$ surjective
- Inverse reversal: $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$

PIGEON-HOLE PRINCIPLE

Statement: If $n > m$, at least one of m holes contains $\lceil n/m \rceil$ or more items

Examples:

- 29 people, 7 days $\rightarrow \lceil 29/7 \rceil = 5$ on same day
- 50 people, 12 months $\rightarrow \lceil 50/12 \rceil = 5$ in same month
- 26 students, 7 cars $\rightarrow \lceil 26/7 \rceil = 4$ per car minimum

PROGRAMMING (PYTHON)

Set Operations

`A.union(B)` # $A \cup B$

`A.intersection(B)` # $A \cap B$

`A.difference(B)` # $A - B$

`A.symmetric_difference(B)` # $A \Delta B$

Check Relation Properties

`is_reflexive = all((a,a) in R for a in A)`

`is_symmetric = all((b,a) in R for (a,b) in R)`

`is_transitive = all((a,c) in R for (a,b) in R if (b,c) in R)`

Function Composition

`def fog(x): return f(g(x))`

Factorial & Permutation

`import math`

`math.factorial(n)` # $n!$

$P(n,r) = n! / (n-r)!$

$C(n,r) = n! / (r!(n-r)!)$

QUICK PROBLEM CHECKLIST

- ☐ Identify if R is a relation (subset of $A \times B$?)

- ☐ Test reflexivity (all (a,a) present?)
- ☐ Test symmetry (reverse pairs exist?)
- ☐ Test transitivity (chain property holds?)
- ☐ Verify function (one output per input?)
- ☐ Check injectivity (different inputs $\rightarrow$ different outputs?)
- ☐ Check surjectivity (range = codomain?)
- ☐ Find inverse (if bijective, reverse the mapping)
- ☐ Compose correctly (apply right function first)
- ☐ Apply ceiling function for Pigeon-Hole ($\lceil n/m \rceil$)

Remember:

- Relations are SUBSETS of Cartesian products
- Functions are SPECIAL relations (1 output per input)
- Equivalence relations PARTITION sets
- Inverse functions exist ONLY for bijective functions
- Composition is evaluated RIGHT-TO-LEFT